Problem Statement:

Given the array nums consisting of 2n elements in the form $[x_1,x_2,...,x_n,y_1,y_2,...,y_n]$.

Return the array in the form $[x_1,y_1,x_2,y_2,...,x_n,y_n]$.

Example 1:

Input: nums = [2,5,1,3,4,7], n = 3

Output: [2,3,5,4,1,7]

Explanation: Since $x_1=2$, $x_2=5$, $x_3=1$, $y_1=3$, $y_2=4$, $y_3=7$ then the answer is [2,3,5,4,1,7].

Example 2:

Input: nums = [1,2,3,4,4,3,2,1], n = 4

Output: [1,4,2,3,3,2,4,1]

Example 3:

Input: nums = [1,1,2,2], n = 2

Output: [1,2,1,2]

Constraints:

- 1 <= n <= 500
- nums.length == 2n
- 1 <= nums[i] <= 10^3

Program:

```
class Solution {
  public int[] shuffle(int[] nums, int n) {
    int[] ans = new int[2 * n];
    int index = 0;

    // Interleave x and y parts
    for (int i = 0; i < n; i++) {
      ans[index++] = nums[i];    // x part
      ans[index++] = nums[i + n]; // y part
    }
```

```
        return ans;
    }

    // Quick test
    public static void main(String[] args) {
        Solution sol = new Solution();

        int[] nums1 = {2,5,1,3,4,7};
        int[] ans1 = sol.shuffle(nums1, 3);
        System.out.println(java.util.Arrays.toString(ans1)); // Output: [2,3,5,4,1,7]

        int[] nums2 = {1,2,3,4,4,3,2,1};
        int[] ans2 = sol.shuffle(nums2, 4);
        System.out.println(java.util.Arrays.toString(ans2)); // Output: [1,4,2,3,3,2,4,1]

        int[] nums3 = {1,1,2,2};
        int[] ans3 = sol.shuffle(nums3, 2);
        System.out.println(java.util.Arrays.toString(ans3)); // Output: [1,2,1,2]
    }
}
```

1. Input: nums = [2,5,1,3,4,7], n = 3

Output: [2,3,5,4,1,7]

2. Input: nums = [1,2,3,4,4,3,2,1], n = 4

Output: [1,4,2,3,3,2,4,1]

3. Input: nums = [1,1,2,2], n = 2

Output: [1,2,1,2]